

Week 1 Topic 1: Linear Equations.

Definition

A linear equation in n variables $x_1, x_2, \dots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are constant real numbers.

*variables are
to the power of one*

Examples

The following are all linear equations, because they can be rearranged into the standard form above, maybe with different symbols for the variables:

$$x + 3y = 7$$

$$x_1 + 2x_2 + 2x_3 + x_4 = x_5$$

$$10p = q + 1$$

$$x_1 + x_2 + \dots + x_n = 1$$

Start of Video 1.1.1

Examples

The following are ^{non-linear} not linear equations:

$$xy = 2$$

$$\sin(q) + \cos(f) = 0.2$$

$$x_1^2 + x_2^2 + \dots + x_n^2 = 1$$

$$x = e^y$$

What is a solution?

Consider the equation

$$x + 3y = 7.$$

The assignment

$$\begin{cases} x = 1 \\ y = 2 \end{cases}$$

is a **solution** to this equation because when we substitute in these values the equation becomes true:

$$(1) + 3 * (2) = 7.$$

The first goal in Linear Algebra is usually to understand the **solution set** of a linear equation.

The solution set of a linear equation is the **collection of all possible solutions** of the equation.

There will be different ways of understanding it, such as *algebraically*, via a *parameterization*, or maybe *geometrically*.

To talk about this all precisely we first need to think about where the solution set would “live”. (i.e. how to give a precise logical definition in terms of some simple set).

Introducing $\mathbb{R}^n$: *the set of all possible n -tuples.*

Fix some integer $n > 0$.

An **n -tuple of real numbers** is just an *ordered* list of exactly n real numbers.

Here are a few examples of, say, 5-tuples of real numbers:

$$(1, 0, 1, 0, 1)$$

$$(\pi, 0, -\pi, 0, 0)$$

$$\left(\frac{1}{2}, 7, 6, -\frac{5}{7}, 0\right)$$



Note that the **ordering is important** so the following two 3-tuples are different:

$$(1, 2, 3) \neq (1, 3, 2).$$

Start of Video 1.1.2

Introducing $\mathbb{R}^n$: the set of all possible n -tuples.

Definition

Let n be an integer $n > 0$.

The set $\mathbb{R}^n$ is defined to be the set of all possible n -tuples of real numbers.

This is probably something you'll think about carefully in *MH1300 Foundations of Mathematics*. The explanation for the notation $\mathbb{R}^n$ is that we are taking the "Cartesian product" of n copies of the set $\mathbb{R}$:

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_n.$$

(This means: An element of $\mathbb{R}^n$ consists of a choice of one element from each one of these n copies of $\mathbb{R}$.)

Definition of the "solution set of a linear equation".

Fix some linear equation.

every possible
solutions

Assume the equation has n variables which are declared in advance and **are ordered**.

(It is possible one of the declared variables doesn't appear explicitly in the equation, in which case we assume it appears with coefficient 0.)

For example, maybe our equation looks like this, with variables ordered so x_1 is first, x_2 is second, etc:

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n = d.$$

A solution $\begin{cases} x_1 = a_1 \\ \vdots \\ x_n = a_n \end{cases}$ corresponds to the element of $\mathbb{R}^n$ you get by

listing the values in the fixed order: $(a_1, \dots, a_n) \in \mathbb{R}^n$.

*(Note: this last step is why we needed to **order the variables!** Otherwise the correspondence is ambiguous!)*

Definition

The **solution set** of the linear equation

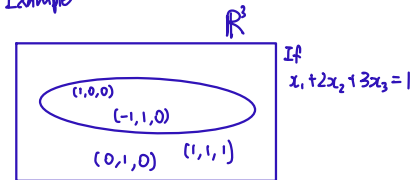
$$c_1x_1 + c_2x_2 + \dots + c_nx_n = d$$

in the variables $x_1, \dots, x_n$ is the subset of $\mathbb{R}^n$

$$S \subset \mathbb{R}^n$$

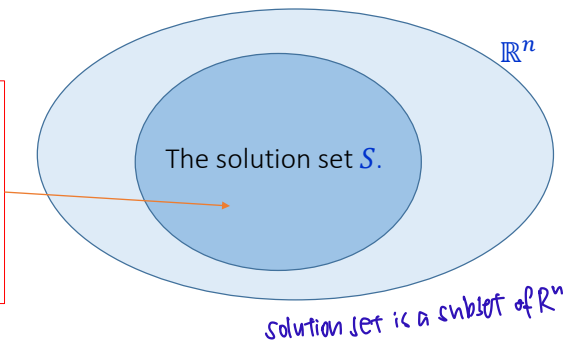
consisting of all n -tuples that correspond to solutions of the equation.

Example:



Illustration

*The most important motivating question of this course is:
How can we compute S and what are its properties inside $\mathbb{R}^n$?*



Comment in passing about set theory notation

On the previous slide I wrote:

$$S \subset \mathbb{R}^n$$

We should be precise about what I mean by this symbol because there are different conventions in mathematics.

To be precise, by $A \subset B$ I will mean exactly the simple definition:

"Every element of A is also an element of B ."

(So this will include the possibility that $A = B$. Some of you may have used $\subseteq$ for this in the past instead of just $\subset$ like we will do in this course. My usage here is the most common meaning in advanced mathematics.)

Video 1.1.3

Exercise

Consider the equation

$$4x_1 - 7x_2 + x_3 = 1$$

and we order the variables in the obvious way x_1, x_2, x_3 .

Let $S \subset \mathbb{R}^3$ denote the solution set of this equation.

Note that $(2, 1, 0)$ is an element of the solution set of the equation because indeed:

$$4 * 2 - 7 * 1 + 0 = 1.$$

Write down another three elements of S .

$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 1 \end{cases}$ <i>is an element of</i> $(0, 0, 1) \in S \rightarrow \text{solution set}$ <i>* can just fix x_1 and x_2 first, then solve for x_3.</i>	$\begin{cases} x_1 = 1 \\ x_2 = 0 \\ x_3 = -3 \end{cases}$ $(1, 0, -3) \in S$	$\begin{cases} x_1 = 9/4 \\ x_2 = 1 \\ x_3 = -1 \end{cases}$ $(9/4, 1, -1) \in S$ <i>* can also just get x_2 and x_3 first, then solve for x_1</i>
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Start of Video 1.1.4

Example: The case of two variables.

Consider a linear equation with 2 variables x_1 and x_2 :

$$ax_1 + bx_2 = c.$$

In this case, a , b and c are unspecified constants (i.e. they're not variables, they stand for fixed numbers) and we assume at least one of a and b is not zero.

We can draw the solution set by plotting a point in the plane $(x_1, x_2) \in \mathbb{R}^2$ for every solution of the equation.

x_1 -coordinate $\swarrow$ $\searrow$ x_2 -coordinate

What do we expect in this case?

Thinking about familiar examples we expect a ? line

Let's play with a few examples of this sort, and get a feeling for what sort of thing we expect to see and for the theory to explain.

Exercise:

Using your intuition, draw the solution sets $S \subset \mathbb{R}^2$ of the following equations in the variables x_1 and x_2 in the Cartesian plane.

1. $5x_1 + 2x_2 = 10.$

2. $1000x_2 = 100.$

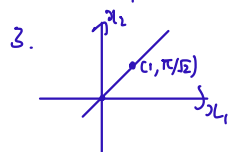
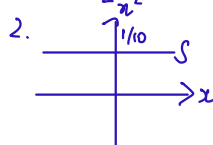
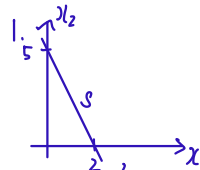
3. $\pi x_1 - \sqrt{2}x_2 = 0.$

4. $0x_1 + 0x_2 = 1.$

5. $0x_1 + 0x_2 = 0.$

if right hand side is zero, it will pass through the origin

special cases
ans is not alive



4 is known as an inconsistent linear eqn, as it has no solⁿ.

4. $S = \emptyset \rightarrow$ no solⁿ

5. $S = \mathbb{R}^2 \rightarrow$ solⁿ is everything

* imaginary numbers are not included in this solution set.
 $\rightarrow$ this question is about real numbers, complex numbers not mentioned
 $\rightarrow$ for q5.

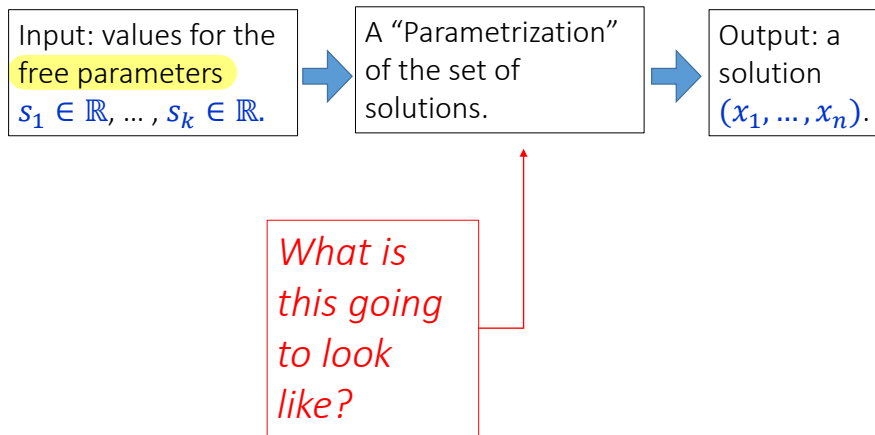
Parameterization of the set of solutions

One nice way to completely describe a set of solutions is to give a *parameterization* for it:

This is where you give a formula for every unknown in your equation in terms of some new variables called *parameters*. Every time you substitute a value into the parameters these formulae give you a solution.

Start of Video 1.1.5

You could think of a parameterization as a machine for generating solutions of the equation:



Definition (Parameterization.)

Consider some linear equation in n variables, such as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

A **parameterization** for the set of solutions of this equation is a collection of n functions *one function for every variable*

$$f_1(s_1, \dots, s_k), \dots, f_n(s_1, \dots, s_k)$$

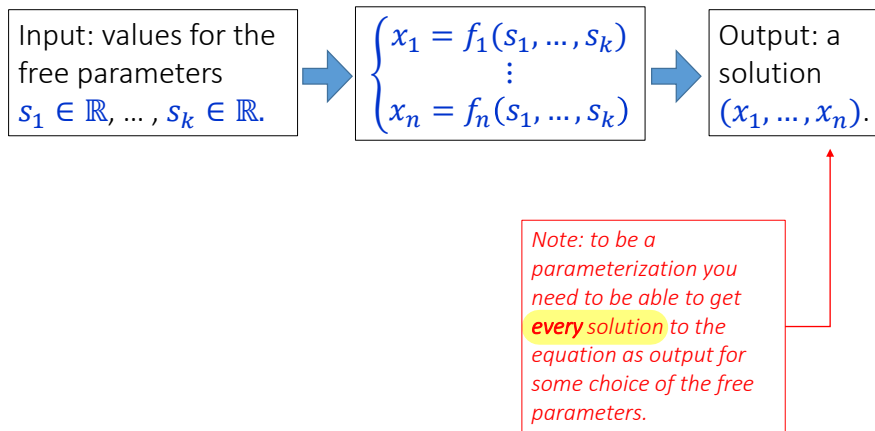
of k variables $s_1, \dots, s_k$ such that whenever we substitute any choice of real numbers into the variables s_i then we get a solution of the equation:

$$\begin{cases} x_1 = f_1(s_1, \dots, s_k) \\ \vdots \\ x_n = f_n(s_1, \dots, s_k) \end{cases}.$$

Moreover we insist that every solution of the equation must arise from some choice of real number for $s_1, \dots, s_k$.

These new variables $s_1, \dots, s_k$ are called **free parameters**.

You could think of a parameterization as a machine for generating solutions of the equation:



Exercise:

Consider the following equations in the variables x_1, x_2 .
 Use a parametrization to describe the complete solution sets of the following equations.

1. $5x_1 + 2x_2 = 10.$
2. $1000x_2 = 100.$
3. $\pi x_1 - \sqrt{2}x_2 = 0.$

Note x_1 doesn't appear explicitly in this equation.
 But we declared at the beginning that the variables are x_1, x_2 .
 So we assume this equation reads: $0x_1 + 1000x_2 = 100.$

Bonus Question:
 $x_1 + 2x_2 + 3x_3 + 4x_4 = 2023$
 as there will be 3 free parameters in the parametrisation of the solution
 let $x_2 = s, x_3 = t, x_4 = u$

$$\begin{cases} x_1 = 2023 - 2s - 3t - 4u \\ x_2 = s \\ x_3 = t \\ x_4 = u \end{cases}$$

Solution to 1:

Consider the given equation: $5x_1 + 2x_2 = 10$.

Note that *whatever* value we set the variable x_2 , we can always solve the resulting equation for x_1 . So we can set x_2 to a new free parameter t , $x_2 = t$, which can take on *any value at all*, and then solve for x_1 in terms of t .

$$5x_1 + 2t = 10 \Rightarrow x_1 = 2 - \frac{2}{5}t.$$

So for any choice of t the formulas $\begin{cases} x_1 = 2 - \frac{2}{5}t \\ x_2 = t \end{cases}$ give a solution to the original equation. *more concrete way of writing the soln*

An equivalent, short way of completely describing the resulting set of solutions is: $S = \left\{ \left(2 - \frac{2}{5}t, t \right) \in \mathbb{R}^2; t \in \mathbb{R} \right\} \subset \mathbb{R}^2$.
presented in vector notation

Let's quickly make sure we all understand this notation I just used:

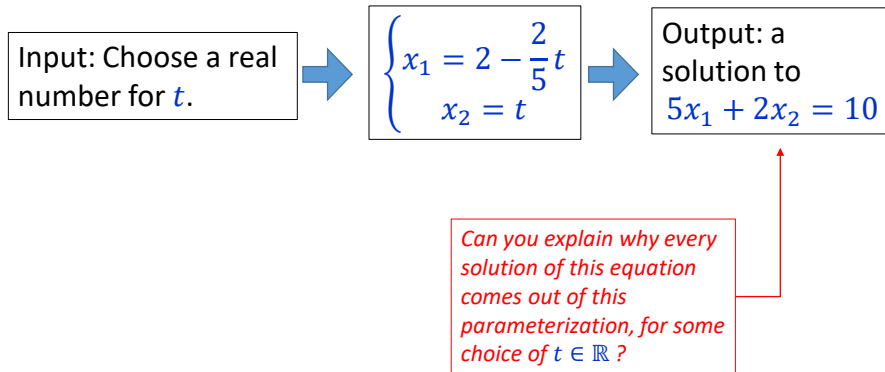
$$S = \left\{ \left(2 - \frac{2}{5}t, t \right) \in \mathbb{R}^2; t \in \mathbb{R} \right\} \subset \mathbb{R}^2.$$

This says: "The set of all possible ordered pairs of numbers in $\mathbb{R}^2$ you can get by substituting some choice of parameter t into the formula $\left(2 - \frac{2}{5}t, t \right)$."

Thus, some examples:

- Substituting $t = 0$ we learn $(2, 0) \in S$.
- Substituting $t = 0.5$ we learn $\left(\frac{9}{5}, \frac{1}{2} \right) \in S$.
- Substituting $t = -10$ we learn $(6, -10) \in S$.
- ETC

Alternatively if we draw this parameterization as a machine:



Exercise: Use a parametrization to describe the complete solution sets of the following equations.

1. $5x_1 + 2x_2 = 10$.
2. $1000x_2 = 100$.
3. $\pi x_1 - \sqrt{2}x_2 = 0$.

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Video 1.1.6

Solution to 2:

We have to make x_1 the free parameter.
If done otherwise, we will not be able to solve since coef. of x_1 is zero.

In this case, we can set x_1 to anything and then solve the equation for x_2 .
We set $x_1 = t$, and then solve:

$$1000x_2 = 100 \Rightarrow x_2 = \frac{1}{10}$$

So the solution set is $\{(t, 1/10) \in \mathbb{R}^2; t \in \mathbb{R}\}$.

Exercise: Use a parametrization to describe the complete solution sets of the following equations.

1. $5x_1 + 2x_2 = 10.$
2. $1000x_2 = 100.$
3. $\pi x_1 - \sqrt{2}x_2 = 0.$

Solution to 3:

In this case, we can set x_2 to a free parameter t and then solve the resulting equation for x_1 .

We set $x_2 = t$, and then solve:

$$\pi x_1 - \sqrt{2}t = 0 \Rightarrow x_1 = \frac{\sqrt{2}}{\pi}t$$

So the solution set is: $\left\{ \left(\frac{\sqrt{2}}{\pi}t, t \right) \in \mathbb{R}^2; t \in \mathbb{R} \right\}$

Exercise: Use a parametrization to describe the complete solution sets of the following equations.

1. $0x_1 + 0x_2 = 1.$
2. $0x_1 + 0x_2 = 0.$

Video 1.1.7

Solution to 1:

In this case there are no solutions, because no matter what we set the variables to, the LHS is always zero. So the set of solutions has **no free parameters**: $S = \emptyset$. (This $\emptyset$ is notation for the **empty set**.)

Solution to 2:

In this case every possible pair (x_1, x_2) solves the equation.

Thus: $S = \mathbb{R}^2$.

We can still express this using a parameterization by using two free parameters: $S = \{(r, s); r, s \in \mathbb{R}\}$.

Week 1 Topic 2: Geometric Vectors

We just described the complete set of solutions to

$$5x_1 + 2x_2 = 10$$

by the parametrization $\left\{\left(2 - \frac{2}{5}t, t\right) \in \mathbb{R}^2; t \in \mathbb{R}\right\}$.

Earlier, when we thought about this set intuitively we realized that this set must form a line in $\mathbb{R}^2$.

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Video 1.2.1

Question: How can we understand the geometry (i.e. the “picture”) of this set directly from this formula?

Answer:

The theory of **Vectors** is a powerful tool for revealing geometry in a problem.

We’ll now learn a few basic things about vectors then come back to think about this set.

Vectors

Next we'll discuss:

- What are vectors?
- What are the two fundamental operations on vectors?
- These two operations can be interpreted geometrically.
- We can use vectors to geometrically interpret sets of solutions to linear equations.

Vectors

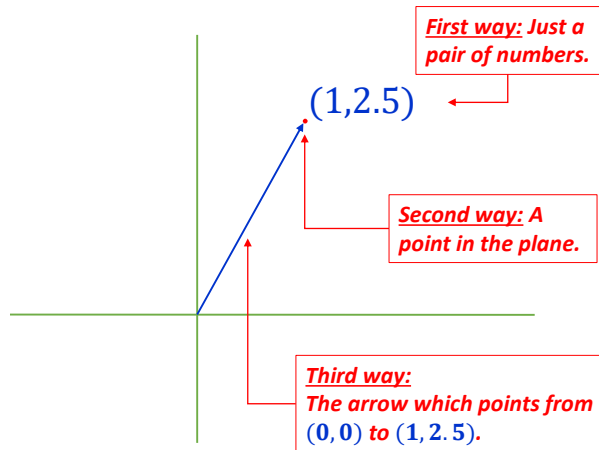
Let's start by thinking about the set $\mathbb{R}^2$. The elements will be called **vectors**.

There are three natural ways of thinking about the points of $\mathbb{R}^2$:

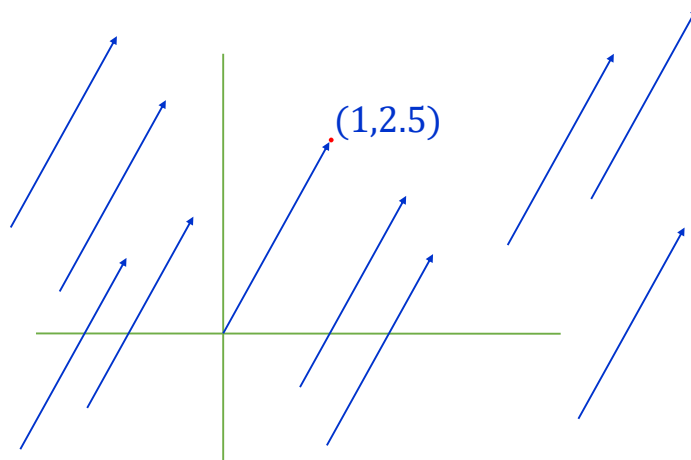
1. Just an ordered list of 2 real numbers (x_1, x_2) . (This is our fundamental definition of $\mathbb{R}^2$.)
2. A point in the plane with coordinates (x_1, x_2) .
3. An arrow from $(0,0)$ to the point (x_1, x_2) . Sometimes we will call this a **geometric vector**, sometimes just vector, sometimes **arrow**.
↓
direction + length

Example:

There are three ways of thinking about the element $(1, 2.5) \in \mathbb{R}^2$.

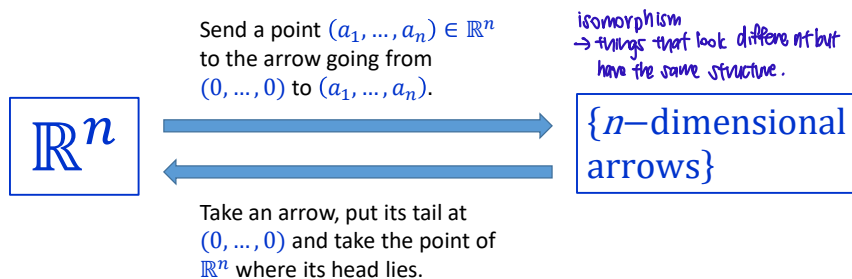
Comment:

A geometric vector (“arrow”) tells you a **length and direction**. Where it starts isn’t part of the information it contains. So we can pick it up and move it around freely and it stays the same, as long as the length and direction are the same. The following are all the same (geometric) vector:



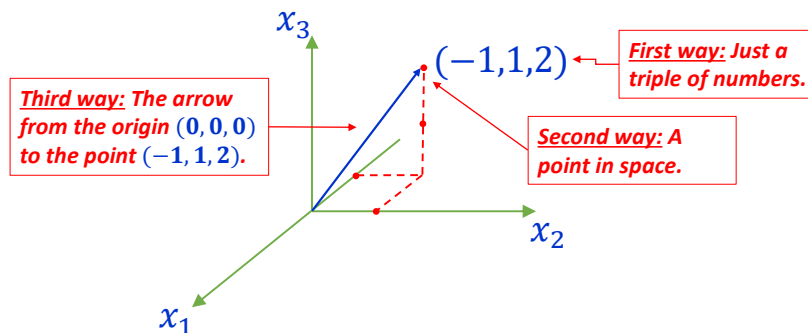
When we say that two different-looking things are “the same” in mathematics, we should give a precise explanation of how the two sets correspond with each other. To do this we should say how to get back and forth between the two sets.

In this case we want think of the following two sets as “the same”. To do that we need to say precisely how an element in the first set corresponds to an element in the second set.



Similarly there are 3 natural ways of thinking about the set $\mathbb{R}^3$:

- Just an ordered list of 3 real numbers (x_1, x_2, x_3) .
- A point in 3-dimensional space with coordinates (x_1, x_2, x_3) .
- An arrow from $(0,0,0)$ to the point (x_1, x_2, x_3) .



$\mathbb{R}^n$

Of course all this generalizes to $\mathbb{R}^n$.

In this case a vector is defined to be a list of n real numbers.

It is hard (or impossible) to visualize “ n -dimensional space” directly but this idea can be useful.

↓
usually only
look at it when
 $n=2$ or 3
for geometric visualization

Notation

When we use symbols to represent vectors we sometimes want to show they are different to ordinary numbers.

In this course we'll usually write a letter with an arrow over the top, like this:

$$\vec{v} \in \mathbb{R}^n.$$

The zero vector

Start of Video 1.2.2

if we think about it as a point in space, it is the origin.

There is a special vector called the **zero vector**.

This is just the element consisting of all zeros:

$$\underbrace{(0, \dots, 0)}_n \in \mathbb{R}^n.$$

We denote it: $\vec{0} \in \mathbb{R}^n$.

↓
zero vector

Equality of vectors

When are two vectors **equal**? If $\vec{v}$ and $\vec{w}$ are vectors, then when can we say $\vec{v} = \vec{w}$?

They are equal precisely when:

1. They live in the same $\mathbb{R}^n$.
2. And the corresponding components in the two vectors are equal.

In other words: When we write out the vectors

$$\vec{v} = (v_1, \dots, v_n) \text{ and } \vec{w} = (w_1, \dots, w_n)$$

then the following equations are **all** true:

$$v_1 = w_1, \dots, v_n = w_n.$$

Operations on Vectors

There are two **fundamental operations** on vectors.

1. We can add two vectors together to make a new one.
2. We can multiply a vector by a real number.

Vocabulary: For historical reasons single real numbers in this situation are sometimes called **scalars**, to distinguish them from **vectors**.

We'll see that we can interpret these operations **geometrically** which is why vectors will be so useful in understanding the geometry in algebraic problems.

Fundamental properties of vector

1. there is a zero vector
2. 2 fundamental operation

Vector operation #1: Adding vectors in $\mathbb{R}^n$.

We add two vectors by *simultaneously* adding up their components:

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

Example:

Here is a simple example in $\mathbb{R}^3$:

$$(1, 2, 3) + (-5, 0, 100) = (-4, 2, 103).$$

Quick question:

Take an arbitrary vector $\vec{v} \in \mathbb{R}^n$.

Then: $\vec{v} + \vec{0} = ?$.

Answer: $\vec{v}$

How to add vectors geometrically:

1. Take two arrows, the first corresponding to the vector $\vec{v}_1$ and the second corresponding to $\vec{v}_2$.

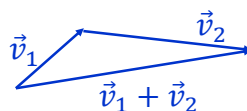


if we can move vectors around, they are not fixed in space.

2. To find the arrow corresponding to $\vec{v}_1 + \vec{v}_2$, pick up the arrow for $\vec{v}_2$ and move it so its tail lies at the head of the arrow for $\vec{v}_1$.



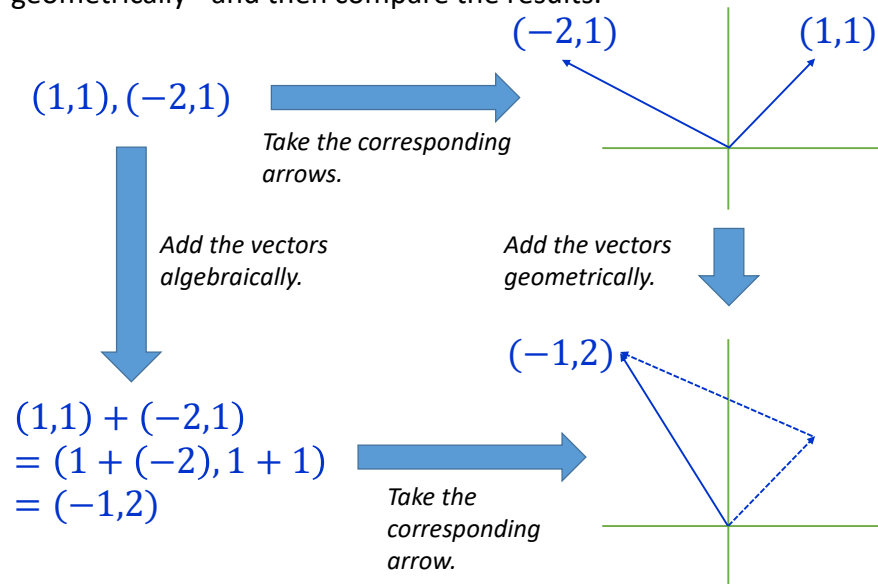
3. The sum $\vec{v}_1 + \vec{v}_2$ is the arrow you now get which goes from the tail of $\vec{v}_1$ to the head of $\vec{v}_2$.



head to tail!

$$\star (1, 1, 0) \neq (1, 1)$$

Example: Consider the two vectors: $(1,1), (-2,1) \in \mathbb{R}^2$.
Let's add them in two different ways - algebraically and geometrically - and then compare the results.



Vector operation #2: Multiplication by a scalar.

We multiply a vector $(a_1, \dots, a_n) \in \mathbb{R}^n$ by a scalar $c \in \mathbb{R}$ by simultaneously multiplying every entry by the scalar

$$c(a_1, \dots, a_n) = (ca_1, \dots, ca_n) \in \mathbb{R}^n.$$

Example:

Here is a simple example in $\mathbb{R}^5$:

$$(-10)(1, -0.2, 2.5, 10, 0) = (-10, 2, -25, -100, 0) \in \mathbb{R}^5.$$

How to multiply by a scalar geometrically:

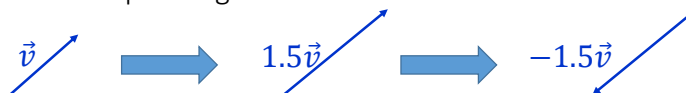
1. Take an arrow corresponding to some vector $\vec{v}$ and a scalar $c \in \mathbb{R}$. For example consider $c = -1.5$ and the following vector:



2. Stretch or squash the vector so that the new vector is parallel to the old vector but has the length stretched or squashed by the factor $|c|$.



3. Finally, if the sign of c is negative **reverse the direction** of the vector otherwise leave it unchanged. The result is the vector corresponding to $c\vec{v}$. remember to change the direction!!



Exercise:

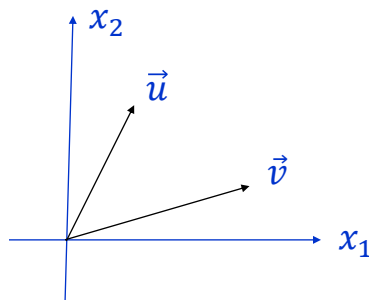
Video 1.2.3

Use the geometric properties of vectors to give a geometric explanation of the difference $\vec{u} - \vec{v}$ of two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$.

$$\begin{array}{c} \vec{u} - \vec{v} \\ \downarrow \\ -\vec{v} = (-1)\vec{v} \end{array}$$

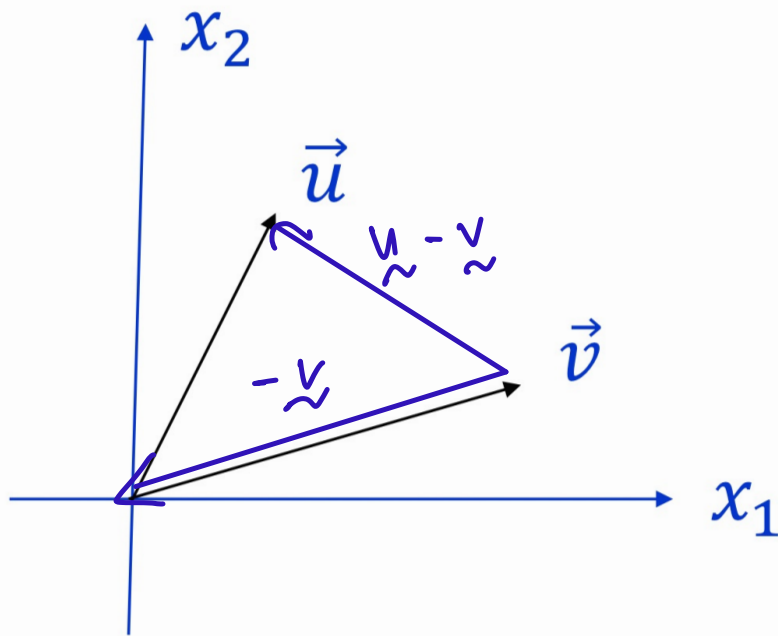
Solution:

Let's consider an example in $\mathbb{R}^2$.



$$\vec{u} - \vec{v} = \vec{u} + (-1)\vec{v}$$

Solution (continued):



Solution (continued):

Place the tails of both vectors at the origin. $\vec{u} - \vec{v}$ is now the vector which points from the head of $\vec{v}$ to the head of $\vec{u}$.

Start of Video 1.2.4

Now let's do some examples of using vectors to decode the geometric meaning of some sets.

Example

Fix a vector $(a, b) \in \mathbb{R}^2$.

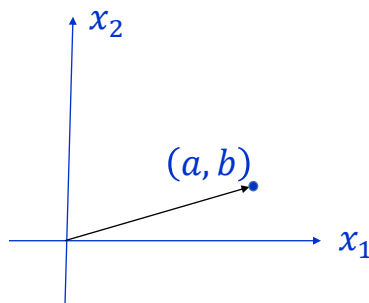
Consider the following set of all multiples of (a, b) :

$$S = \{t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}. \quad \text{t is the variable!}$$

Describe the resulting subset $S \subset \mathbb{R}^2$.

Discussion:

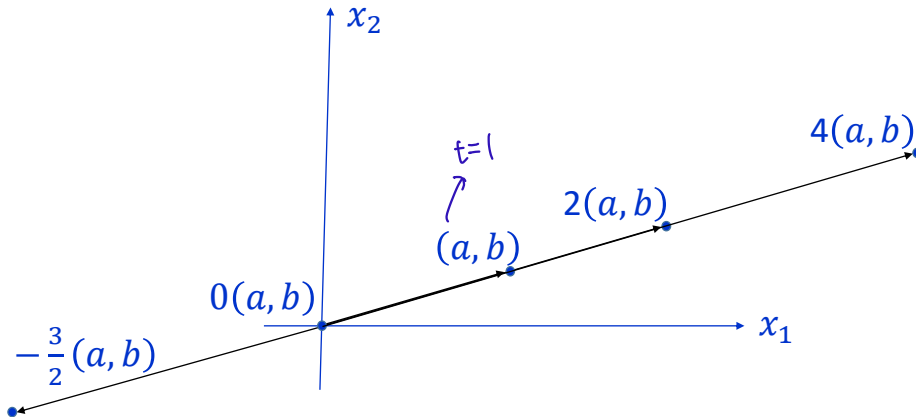
The point $(a, b) \in \mathbb{R}^2$ is the endpoint of an arrow from the origin.



Discussion (continued):

When we take a multiple $t(a, b)$ the geometric operation is to take an arrow parallel to the original arrow (a, b) but which has been stretched by a factor t . We put the resulting arrow at the origin again and take the endpoint.

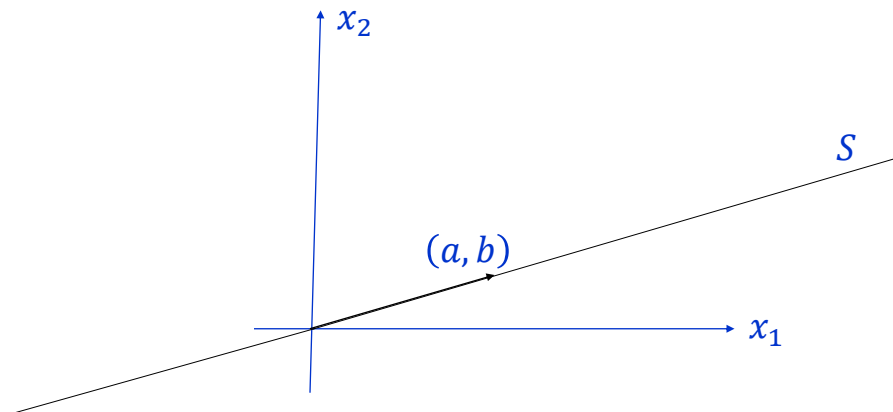
Here are a few points from the set $\{t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}$.

Discussion (Conclusion):

Thus:



The full set $S = \{t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}$ is the line through the origin parallel to the vector (a, b) .



Example

Start of Video 1.2.5

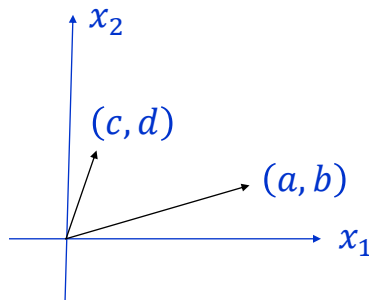
Now let's make this example a bit more interesting by including a second vector (c, d) . Consider the following set:

$$S' = \{(c, d) + t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}.$$

Describe the resulting subset $S' \subset \mathbb{R}^2$.

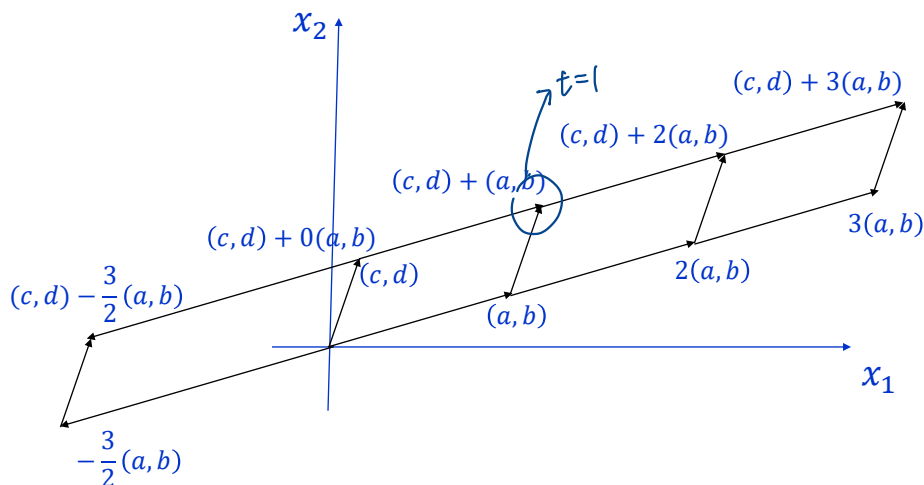
Discussion:

Consider the two vectors:



We can understand where the points $(c, d) + t(a, b)$ lie by adding the two terms $t(a, b)$ and (c, d) using the parallelogram law.

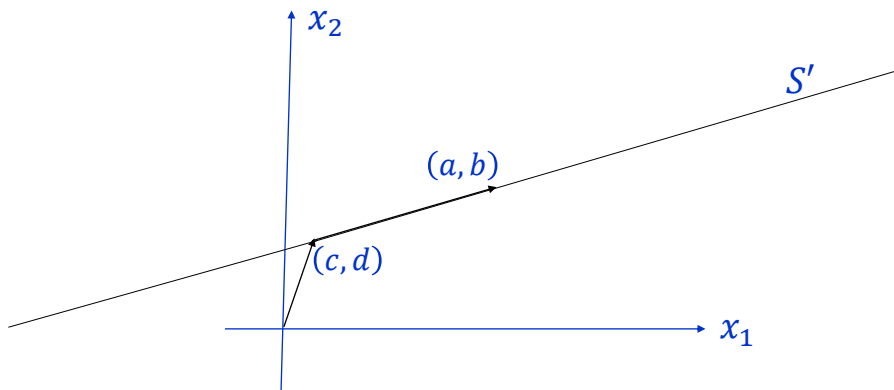
To understand this let's draw a few cases $t = 1, 2, 3, -3/2, 0$.



Discussion (Conclusion):

Thus:

The full set $S' = \{(c, d) + t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}$ is the line through the point (c, d) parallel to the vector (a, b) .

Comment

The previous example was a special case of a simple geometric property. (Explored in this week's tutorials.)

Let S be some set of vectors $S \subset \mathbb{R}^n$ and fix a vector $\vec{w} \in \mathbb{R}^n$.

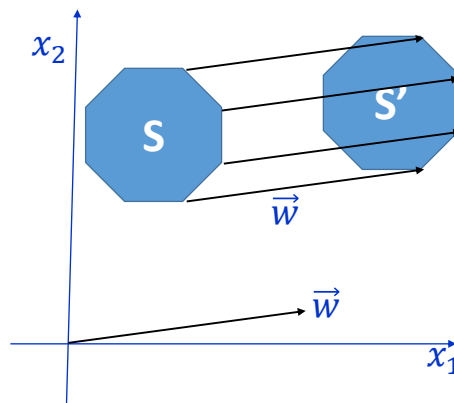
Build a new subset as follows: $S' = \{\vec{v} + \vec{w}; \vec{v} \in S\} \subset \mathbb{R}^n$.

Question

What is the geometric relationship between S and S' ?

Answer

S' is obtained by shifting (we say "translating") S by $\vec{w}$.



Comment (continued).

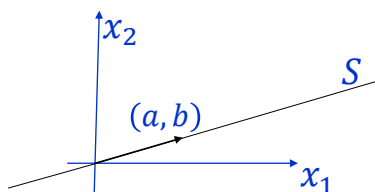
So we could have understood the previous example

$$S' = \{(c, d) + t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}.$$

in a different way as follows:

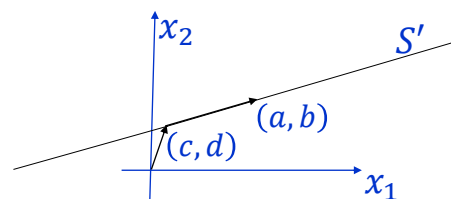
First understand

$$S = \{t(a, b) \in \mathbb{R}^2; t \in \mathbb{R}\}$$



Then shift that set S by

(c, d) to get the requested set S' .



Example

Start of Video 1.2.6

Earlier we described the complete set of solutions to the equation

$5x_1 + 2x_2 = 10$ by the parametrization

$$\left\{ \left(2 - \frac{2}{5}t, t \right) \in \mathbb{R}^2; t \in \mathbb{R} \right\}.$$

Let's use vectors to understand what this set looks like.

Geometric deductions:

First we'll rewrite this formula for a point so we can see explicitly how it is built from geometric operations:

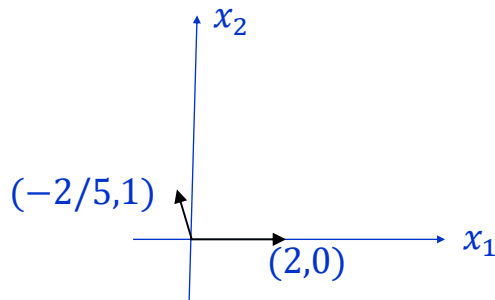
$$\left(2 - \frac{2}{5}t, t \right) = (2, 0) + \left(-\frac{2}{5}t, t \right) = (2, 0) + t \left(-\frac{2}{5}, 1 \right)$$

Thus: from the earlier examples we expect this set of solutions to be the straight line going through the point $(2, 0)$ in the direction of the vector $(-2/5, 1)$.

Let's understand again why that is true:

The formula we are studying is: $(2,0) + t\left(-\frac{2}{5}, 1\right)$.

We'll start by plotting the basic building blocks $(2,0)$ and $\left(-\frac{2}{5}, 1\right)$ and build up from there:



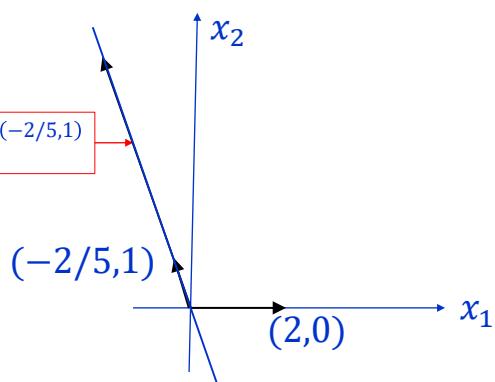
Next question: What do the points $t\left(-\frac{2}{5}, 1\right)$ look like on this picture?

Geometric deductions (continued):

The formula we are studying is: $(2,0) + t\left(-\frac{2}{5}, 1\right)$.

Next let's draw the vectors $t\left(-\frac{2}{5}, 1\right)$. For different $t \in \mathbb{R}$ we get different vectors, but they are all parallel to $\left(-\frac{2}{5}, 1\right)$. If we take all possibilities for all t we get a line through the origin parallel to $(-2/5, 1)$.

E.g. this is the vector $t(-2/5, 1)$
for the choice $t = 5$.

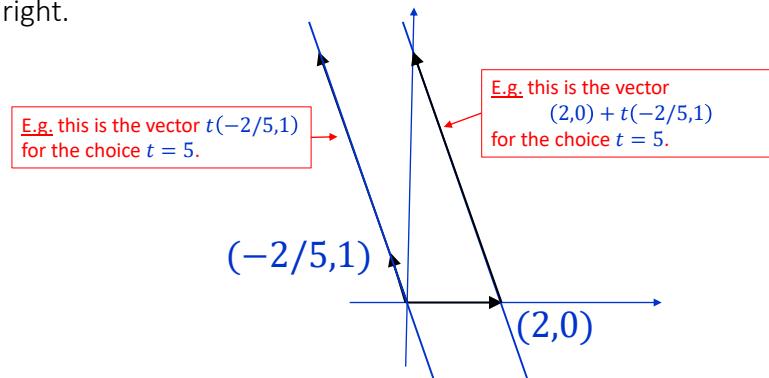


Geometric deductions (continued):

The formula we are studying is: $(2,0) + t\left(-\frac{2}{5}, 1\right)$.

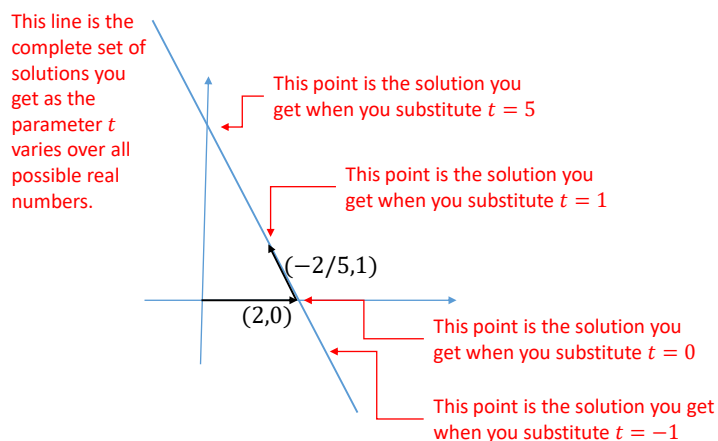
To finish: we have to add these two pieces together. Remember how to add vectors: move one of the vectors so the tail of it lies at the head of the first vector.

The term $t(-2/5, 1)$ gives the line through the origin in the direction $(-2/5, 1)$. Adding $(2,0)$ to it shifts every point on the line by 2 units to the right.

Summary picture:

We just described the complete set of solutions to $5x_1 + 2x_2 = 10$ by the formula $\left\{\left(2 - \frac{2}{5}t, t\right) \in \mathbb{R}^2 : t \in \mathbb{R}\right\}$.

Using vectors we drew this set as follows:



Start of Video 1.2.7

The case of linear equations in three variables

Consider a linear equation

$$ax + by + cz = d$$

where a, b, c, d are constants and at least one of a, b, c is non-zero.

If we plot a point (x, y, z) in three-dimensional space (denoted $\mathbb{R}^3$) for every solution of this equation then we always get a ? plane.

Example

Consider the linear equation $x + y + z = 1$.

1. Using intuition, draw a rough diagram of the set of solutions.
2. Use a parametrization to completely describe the set of solutions. (Hint: in this case you'll need two parameters.)
3. Using the parameterization, apply vectors to observe that the solution set forms a plane in $\mathbb{R}^3$.

Solution to 1:

One way to get a feel for the set of solutions is to look at the intersections with the coordinate planes.

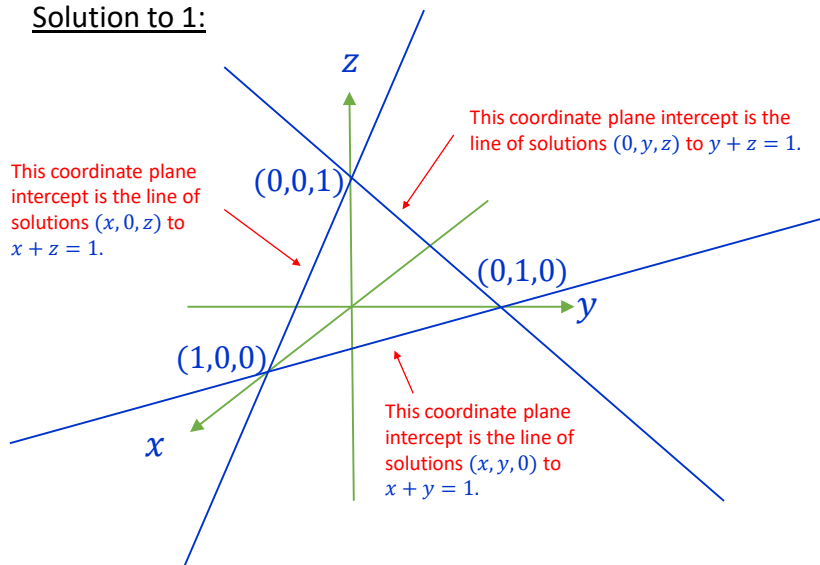
To begin: The intercept with the x - y plane is exactly the set of solutions with $z = 0$. Substituting $z = 0$ into $x + y + z = 1$ we obtain the equation $x + y = 1$.

Doing this for all three coordinate planes we deduce the intersections are:

- In the x - y plane: $x + y = 1. \rightarrow z=0$
- In the x - z plane: $x + z = 1. \rightarrow y=0$
- In the y - z plane: $y + z = 1. \rightarrow x=0$

These are all straight lines.

We guess that the set of solutions to $x + y + z = 1$ is the plane containing these three lines.

Solution to 1:

Solution to 2:

Note that we can set y and z to *anything at all*, and there will be *exactly one* choice of x which then solves the equation.

So we introduce free parameters s and t , set $y = s$ and $z = t$ and solve for x :

$$x + y + z = 1 \Rightarrow x = 1 - s - t.$$

Interpretation: For every $s, t \in \mathbb{R}$ the assignments

$$\begin{cases} x = 1 - s - t \\ y = s \\ z = t \end{cases} \text{ yield a solution to the given equation.}$$

Moreover, because x is uniquely determined by y and z , every solution is obtained from some s and t .

Using set theory notation we can denote the solution set by:
 $\{(1 - s - t, s, t) \in \mathbb{R}^3 : s, t \in \mathbb{R}\}.$

Solution to 3:

Start of Video 1.2.8

The solution set we found is:

$$\{(1 - s - t, s, t) \in \mathbb{R}^3 : \underbrace{s, t}_{\text{free parameters}} \in \mathbb{R}\}$$

To help visualize this set we can rewrite this expression in vector notation and “separate the variables”:

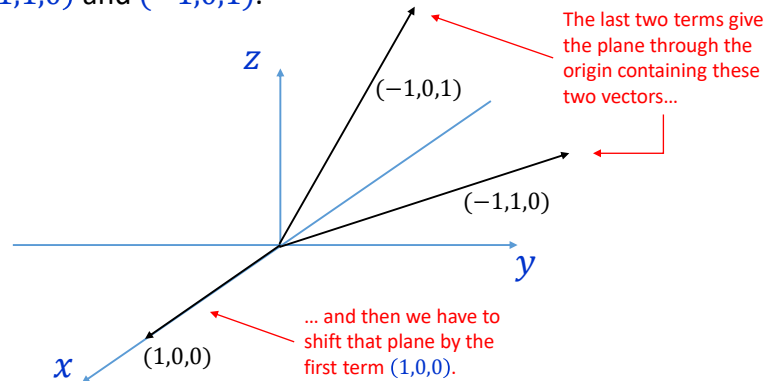
$$\begin{aligned} &= (1, 0, 0) + (-s - t, s, t) \\ &= (1, 0, 0) + (-s, s, 0) + (-t, 0, t) \\ &= (1, 0, 0) + s(-1, 1, 0) + t(-1, 0, 1) \end{aligned} \quad \left\{ (1, 0, 0) + s(-1, 1, 0) + t(-1, 0, 1) \in \mathbb{R}^3 : s, t \in \mathbb{R} \right\}.$$

Like the earlier example, one approach is to understand what we get from this part first. We can then take our answer to that and translate it by the constant vector $(1, 0, 0)$.

$$\{(1,0,0) + s(-1,1,0) + t(-1,0,1) \in \mathbb{R}^3 : s, t \in \mathbb{R}\}.$$

In words, this part is: “All possible points you get to from the origin by going parallel to $(-1,1,0)$ for some distance, followed by parallel to $(-1,0,1)$ for some distance”.

This shape is exactly the plane through the origin containing the two vectors $(-1,1,0)$ and $(-1,0,1)$.



Solution to 3:

In summary: The solution set we found is:

$$\{(1 - s - t, s, t) \in \mathbb{R}^3 : s, t \in \mathbb{R}\}$$

We rewrote it:

$$\{(1,0,0) + s(-1,1,0) + t(-1,0,1) \in \mathbb{R}^3 : s, t \in \mathbb{R}\}.$$

The resulting plane looks like:

